

Andrew Fleck

West Lafayette, IN · (812) 634-7214 · afleck18@gmail.com · linkedin.com/in/andrewfleck

Research + Engineering Experience

Research Assistant – Verma Lab (PI: Prof. Mohit Verma)

Oct 2023 – Dec 2025

Purdue University

- Developed **vision-based sensing pipelines** for state estimation using YOLOv8 instance segmentation and ByteTrack multi-object tracking (Kalman filtering, Hungarian assignment, Mahalanobis gating) for real-time detection and tracking from RGB and thermal video; implemented filtering methods for noise reduction in environmental sensing data
- Developed learning-augmented state estimation frameworks integrating learned perception with probabilistic trajectory estimation; solved inverse imaging and signal reconstruction problems under partial observability via regularized estimation and convex optimization
- Built computational imaging and regression pipelines for environmental sensing using satellite, GDAS, and NWS data; applied supervised learning models (Random Forest, XGBoost) with SHAP-based interpretability for contaminant prediction
- Analyzed how measurement sensitivity affects estimator stability, robustness, and failure conditions; deployed real-time data and inference pipelines on secure cloud platforms (Azure) for monitoring and decision support in resource-constrained environments.

Affiliated Researcher – Purdue University

Dec 2025 – Present

(Independent research collaboration)

- Proved that learned measurement Jacobians degrade contraction in nonlinear observers; introduced representation-aware gain normalization restoring stability without retraining, validated across terrestrial and underwater sensing regimes with time-varying distortion and illumination variability
- Developing a control-theoretic framework for stability assessment of partially observed, forced dynamical systems via time-varying operator models $A(t)$; enables detection of instability-prone regimes beyond statistical baselines
- Formulating state estimation for non-stationary, forced geophysical systems as a convex, physics-constrained inverse problem with time-varying observation operators and exogenous inputs; enabling stable reconstruction under sparse sensing
- Built real-time geospatial data pipelines integrating NOAA, NASA, and USGS sources for streaming ingestion and preprocessing, supporting estimation, stability analysis, and decision-making under dynamically varying sensing conditions

Publications

Geometry-Induced Contraction Degradation and Stabilization of Learning-Enabled Observers

IEEE CDC 2026

(submitted)

State Estimation in Non-Stationary Geophysical Systems under Forcing and Sparse Sensing

In preparation

Operator-Based Stability Certification for Partially Observed, Forced Geophysical Systems

In preparation

Conference Presentations

Spatiotemporal State Estimation and Multi-Object Tracking in Hybrid Biological Systems

Purdue ABE IRS, 2024

Vision-based tracking and probabilistic state estimation for recovering latent dynamics from noisy video data

Project Experience

Robust Estimation Under Partial Observability (Research Project)

- Designed a controlled experimental framework to evaluate how sensing degradation influences estimator reliability, uncertainty characterization, stability assessment, and risk under partial observability.
- Implemented nonlinear state estimation using an Extended Kalman Filter (EKF) under nominal, geometry-dependent, and intermittent sensing regimes while holding the underlying system dynamics fixed to isolate the effects of information loss.
- Developed complementary assessment metrics comparing covariance-derived uncertainty, residual disagreement, observation-derived stability estimates, and risk indicators to characterize estimator reliability under evolving sensing conditions.
- Demonstrated that information loss causes divergence between covariance-derived confidence and realized estimation performance, revealing limitations of covariance-based reliability metrics under controlled sensing degradation.

Senior Thesis – ECG Waveform Reconstruction as an Inverse Imaging Problem

- Reconstructed ECG signals from scanned images under noise and distortion
- Developed image-processing and filtering pipeline preserving physiological dynamics

Engineering Experience

System Engineer – Epic Systems

Dec 2022 – Jul 2023

- Led components of distributed clinical infrastructure deployment (1M patients), emphasizing availability and fault tolerance. Implemented enterprise systems (Azure, Citrix) with secure, compliant data handling
- Built PowerShell automation reducing configuration variability. Designed monitoring and incident-response workflows to manage failure and minimize downtime

Technical Skills

Estimation & Signal Processing: State Estimation, Nonlinear Observers, Recursive Filtering

Control & Systems: Nonlinear Systems, Stability Analysis, Contraction Theory

ML & Sensing: Computer Vision (YOLO-based detection, tracking), Supervised/Unsupervised Learning (RF, XGBoost), Learning-enabled state estimation

Programming: Python (NumPy, SciPy, OpenCV), MATLAB, C, R, SQL, JavaScript

Systems/Hardware: Data Pipelines, Distributed Systems, Azure, Citrix, PowerShell

Education

Purdue University

B.S. Biomedical Engineering, 2022 · Minor in Chemistry (Physical Chemistry, Thermodynamics) · GPA: 3.57/4.0